Design Patterns - Lab 3
Visitor

Goal
Introduce the basic concept of the Visitor pattern and apply the pattern to our book in order to render the book, extract different type of statistics, create the table of content etc.

Tasks

1. Implement Book statistics in order to count how many paragraphs, how many images and how many tables we have.,
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   ……

2. What are the challenges in implementation of this feature?

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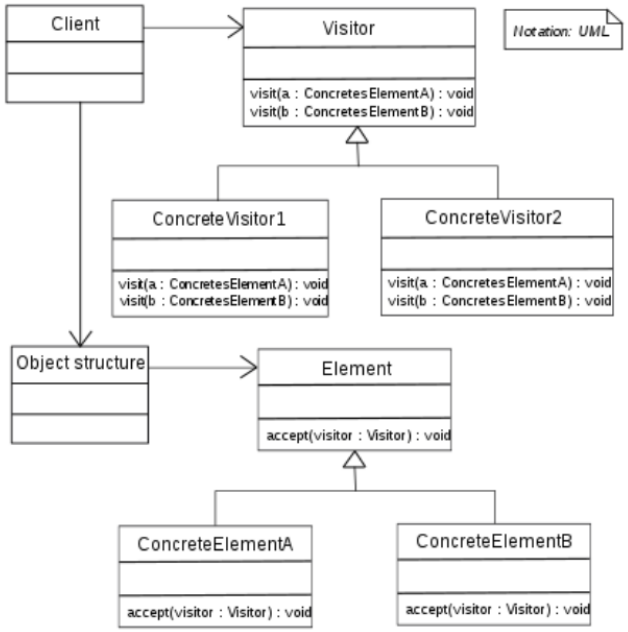
3. The print() method depends on our model classes. How can we decouple this?

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4. What kind of design pattern can we use in order to apply some general processing, like rendering, against a data structure?

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   ……

5. The diagram of the Visitor pattern is:



6. Main purpose of Visitor: perform operations on elements of an object structure. When you need to define new operation you will not need to touch the existing classes you will just need to implement your operation and use it. Will this fit with our current design?

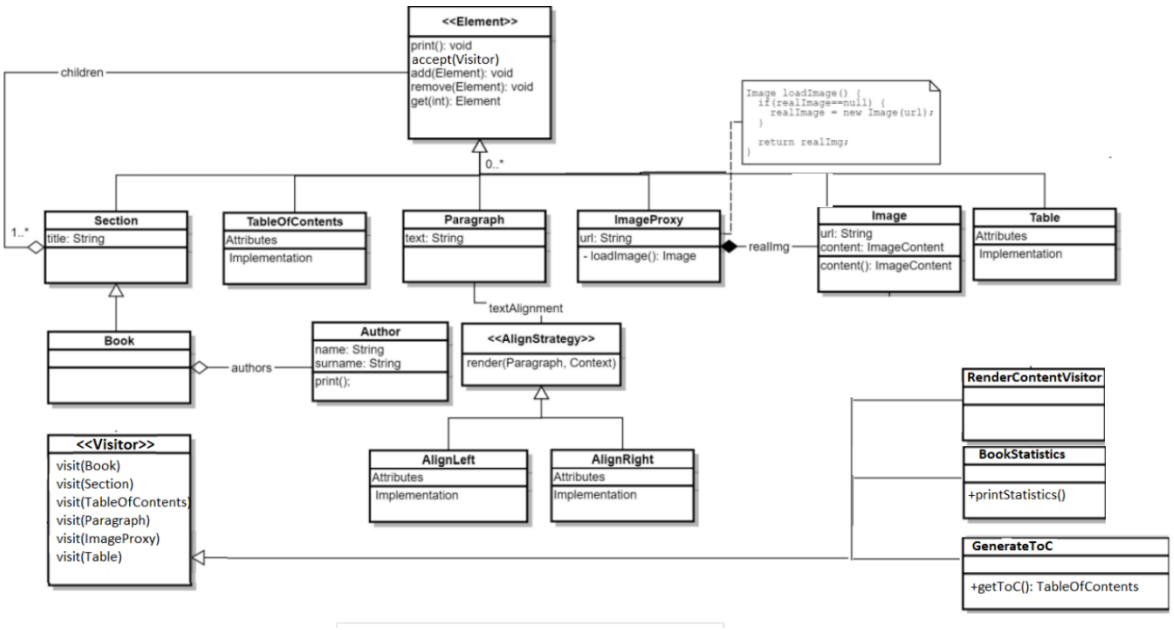
   ……

7. What changes are necessary to our classes in order to implement a render visitor?

   ……

   ……

8. Considering the following class view implement the missing classes and methods



9. If you have trouble implementing you can follow this steps:
    a. Create the Visitor interface having methods for each type of Element (e.g. visit(Book), visit(Paragraph))
    b. Define a new function in Element interface: accept(Visitor visitor)
    c. Implement accept method in each type of Element (pay attention to Section Element)
    d. Implement a concrete Visitor: RenderContentVisitor to render the content; visit(Xxx) will probably be a call to Xxx.render() method
    e. Remove print() from Element and its implementations and implement render() method to print only the necessary information for each element.
    f. Use the RenderContentVisitor in main

```
myBook.accept(new RenderContentVisitor());
```

10. Review the design
    a. What are the advantages and disadvantages of using Visitor?
    b. Analyse the dependency matrix after this refactoring

11. Usage 2: We would like to have a clear statistic on how many paragraphs, images, tables he has in the book. Can we get these statistics in a simple and elegant way?
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    ……...

12. Add the main function to your implementation and add the missing implementation in order to compile and execute the main function code:

```
public static void main(String[] args) throws Exception {
	Section cap1 = new Section("Capitolul 1");
	Paragraph p1 = new Paragraph("Paragraph 1");
	cap1.add(p1);
	Paragraph p2 = new Paragraph("Paragraph 2");
	cap1.add(p2);
	Paragraph p3 = new Paragraph("Paragraph 3");
	cap1.add(p3);
	Paragraph p4 = new Paragraph("Paragraph 4");
	cap1.add(p4);
	cap1.add(new ImageProxy("ImageOne"));
	cap1.add(new Image("ImageTwo"));
	cap1.add(new Paragraph("Some text"));
	cap1.add(new Table("Table 1"));

	BookStatistics stats = new BookStatistics();
	cap1.accept(stats);
	stats.printStatistics();
}
```

13. Expected result

```
Book Statistics:
*** Number of images: 2
*** Number of tables: 1
*** Number of paragraphs: 5
```

14. **Generate Table of Content**